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SUPERVISED LEARNING



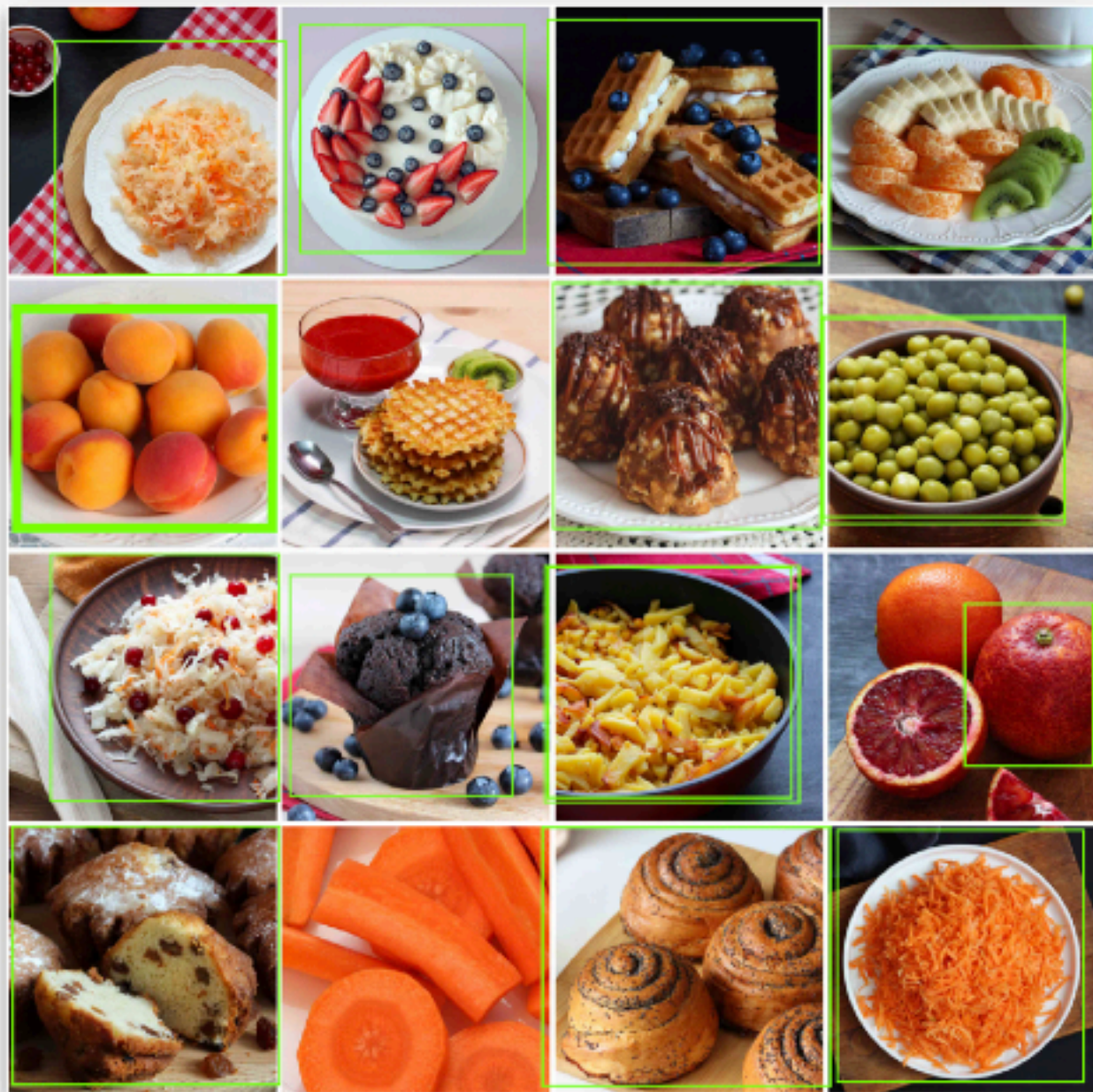
What Kind of Data Types?

The dataset is a collection of tuples:

$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})\}$ Where:

- ▶ $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \dots, x_m^{(i)}]$ is the numerical representation of the image for classification. It is created by flattening pixel values into a 1D array. For instance, a 100x100 pixel image yields a 10,000-element grayscale vector or a 30,000-element RGB vector.
- ▶ $y =$ The bounding box of objects.

Example: Detecting Faces from Images.



Tableware 84%

Bowl 79%

Food 79%

Tableware 78%

Food 78%

Food 78%

Food 77%

Tableware 76%

Food 76%

Food 76%

Object Detection

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- ▶ $y =$ The bounding box of objects.

Example: Detecting Food from Images.



Object Detection

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What Kind of Data Types?

The dataset is a collection of tuples:

$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})\}$ Where:

- ▶ $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \dots, x_m^{(i)}]$ is a sentence or a sequence of words in the source language.
- ▶ y = the translation of the sentence x in the target language

Example: Translating public health advisories or medical guidelines into different languages.

